ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE B · MARKING GUIDELINE

MARKS: 80

Marking guideline for the Academic Central NC(V) Level 4 Electronics and Digital Concepts for Robotics paper 2 · practice b.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, equivalent designs and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 A ✓ 1.1.2 B ✓ 1.1.3 A ✓ 1.1.4 C ✓ 1.1.5 A ✓ 1.1.6 D ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 B ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 C ✓ 1.2.2 G ✓ 1.2.3 J ✓ 1.2.4 B ✓ 1.2.5 E ✓ 1.2.6 H ✓ 1.2.7 D ✓ 1.2.8 A ✓ 1.2.9 I ✓ 1.2.10 F ✓ (10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 2

2.1 Battery → fuse ✓ → 12 V motor rail to the H-bridges ✓; buck converter / regulator → 5 V rail to the sensors ✓; 3,3 V regulator (from 5 V) → ESP32 ✓; common ground and labels ✓ (5)

2.2.1 $P = (V_{in} - V_{out}) \times I$ ✓ $= (12 - 5) \times 0{,}8 = 5{,}6\ W$ ✓ (2)

2.2.2 $\eta = V_{out} \div V_{in} \times 100$ ✓ $= 5 \div 12 \times 100 = 41{,}67\%$ ✓ (2)

2.2.3 A buck converter switches instead of burning off the extra voltage as heat ✓, so it is about 85–95% efficient, runs cool and makes the battery last much longer ✓. (2)

2.3 $P_{out} = 5 \times 0{,}8 = 4\ W$; $P_{in} = 4 \div 0{,}9 = 4{,}44\ W$ ✓✓; $I_{in} = 4{,}44 \div 12 = 0{,}37\ A$ ✓ (3)

2.4 $I = 3 \times 0{,}6 + 0{,}4 = 2{,}2\ A$ ✓; $t = 5 \div 2{,}2$ ✓ $= 2{,}27\ h$ ✓ (3)

2.5 Motors and switching ICs cause fast current changes and voltage spikes / noise on the supply ✓; a capacitor close to each IC supplies these short bursts of current ✓ and keeps the IC's supply steady so it does not reset or give errors ✓. (3)

[20]

QUESTION 3

3.1 Four switches (transistors) form an H around the motor ✓. Forward: one top-left and bottom-right pair conducts ✓ so current flows one way through the motor ✓; reverse: the other diagonal pair conducts and the current reverses ✓. If both switches on one side are on, the supply is shorted (shoot-through), which destroys the switches ✓. (5)

3.2 $V_{avg} = 0{,}75 \times 12$ ✓ $= 9\ V$ ✓ (2)

3.3.1 360×30 ✓ $= 10\,800$ pulses ✓ (2)

3.3.2 $360° \div 10\,800$ ✓ $= 0{,}033°$ ✓ (2)

3.4 $V = 5 \times 3{,}3 \div (1{,}8 + 3{,}3)$ ✓✓ $= 3{,}24$ V ✓ (3)

3.5 $I = (2{,}87 - 2{,}5) \div 0{,}185$ ✓✓ $= 2$ A ✓ (3)

3.6 Encoder pulses are very fast and short ✓; if the main loop is busy (e.g. calculations, communication) it can miss pulses and the position is wrong ✓; an interrupt runs immediately on every pulse edge, so every pulse is counted ✓. (3)

[20]

QUESTION 4

4.1.1 $G = O \cdot P \cdot \bar{F}$ ✓✓ (2)

4.1.2 NOT gate on F ✓; 3-input AND gate (or two 2-input AND gates) ✓ with inputs O, P and $\bar{F}$ ✓; output G labelled, neat ✓ (4)

4.2 Branch: (O AND P) in parallel with a seal-in contact G ✓✓; in series with NC R ✓ and NC F (fault) ✓; driving coil G ✓, e.g. (5)

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|--+--[ O ]--[ P ]--+--[/ R ]--[/ F ]--( G )--|
|  |                |
|  +-----[ G ]------+
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4.3 Rung 1: a part-placed sensor (or gripper-open signal) drives the count input of CTU C1 with preset 10 ✓✓; RESET button drives the reset input of C1 ✓; rung 2: the counter done bit (C1.DN / count ≥ 10) switches on the "Box full" lamp ✓; the lamp goes off when the counter is reset ✓. (5)

4.4 A timer that the program must keep resetting ✓; if the program hangs or crashes, the watchdog times out and resets the controller / stops the motors safely ✓. (2)

4.5 A program can crash, freeze or have bugs ✓; a hard-wired emergency stop cuts the power to the motors directly, so it always works whatever the software is doing ✓. (2)

[20]

TOTAL SECTION B: [60]

TOTAL: 80

END OF MARKING GUIDELINE